



SPIRIT VILLAGE



MISTY PINEWOOD



ECHO CAVERNS



FROSTBLOOM HIGHLANDS

GAME DEVELOPMENT · HUMAN–COMPUTER INTERACTION · DATA SCIENCE

CALMQUEST

The Lost Light

An open-world 2D cozy game designed to help people unwind after work, built around a **transparent, rule-based adaptive comfort system**, privacy-by-design data collection and a reproducible research pipeline, ready for a controlled study with human participants.

9 regions
2 isles, 2 caves

3 acts
story and finale

1,437
automated tests & checks

EN · ID
bilingual, accessible

Author
Dul Kipli Kurniawan

Role
Solo developer & researcher

Source code
github.com/dkiplikurniawan/calmquest

01 · AT A GLANCE

A restful game that is honest about how it adapts to you

Work stress is common, and many people turn to games to switch off. Yet most "relaxation" apps either measure people without explaining how, or rely on black-box "AI" that claims to know how someone feels. CALMQUEST takes the opposite path: **fun first, transparent always**.

10,240×6,144
pixel open world, seeded & streamed in chunks

9 + 4
regions, plus 2 isles and 2 handcrafted caves

3 acts
story with 5 Hearthstones in any order and a finale

2
languages, English and Bahasa Indonesia, switchable live

188
GDScript unit tests on pure, scene-free logic

1,249
automated scenario checks (smoke, overworld, UI)

18 / 18
quality checks passing in the latest full run

0
lines of network code: nothing leaves the device

The problem

Relaxation tools rarely explain what they adapt, why, or what they store. Players cannot verify claims, and researchers cannot reproduce results. Games that could help people rest often add pressure through timers, streaks and failure.

The approach

A complete game with **no fail states**, where every adaptation is a **named, documented, unit-tested rule**, every research datum requires **explicit consent**, and a hidden **Study Mode** turns the same build into a research instrument.

What I built

- A full Godot 4 game: exploration, crafting, farming, fishing, cooking, villagers, puzzles
- An explainable adaptive comfort engine (Gentle / Balanced / Playful)
- A well-being layer: mood check, 12 rest spots, "Breathe with Lumi"
- A consent-based data layer, Study Mode and a tested Python analysis pipeline
- An automated quality suite that plays the whole story end-to-end

COMPLETE

Development · v0.4.0

Feature-complete game, research tooling and documentation, all automated checks passing.

NEXT

Human evaluation

A usability pilot, then a randomized crossover study with 30 working adults (Section 07).

THEN

Public release

Steam (PC) first, followed by Android and iOS.

02 · THE EXPERIENCE

A world that brightens because you were kind to it

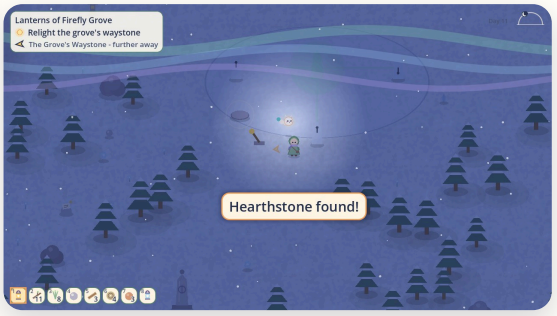
The light of the Lumeria Isles has faded. As a quiet Traveler, guided by a small light spirit named Lumi, the player restores a great Heart Tree and steps out into glowing groves, misty pinewoods, singing caves, a sleepy lighthouse and frosty highlands under an aurora.



The spirit village grows as the world is restored: five villagers with homes, routines and personal stories.



Act 2: relighting the lighthouse at night.



Frostbloom Highlands under the aurora.

> *"There are no timers, no lives, no streaks, no damage and no punishment for stopping. You can't lose. The only thing that changes is how bright the world becomes."*

Kindness as mechanics

Shadow creatures are comforted with light, song or food, never fought. Crops never die, and rain waters them.

Curiosity over checklists

40 recipes are discovered through notes and experiments. Six tools open new parts of the world.

Puzzles without walls

Every puzzle has a reset lever and a three-step hint ladder. Quests can never become impossible.

Accessible by default

Keyboard, mouse and gamepad, remapping, text size, high contrast, reduce motion and captions.

A home of your own

Restore a cottage in 8 steps and decorate it on a grid, inside and out, with 18 decorations.

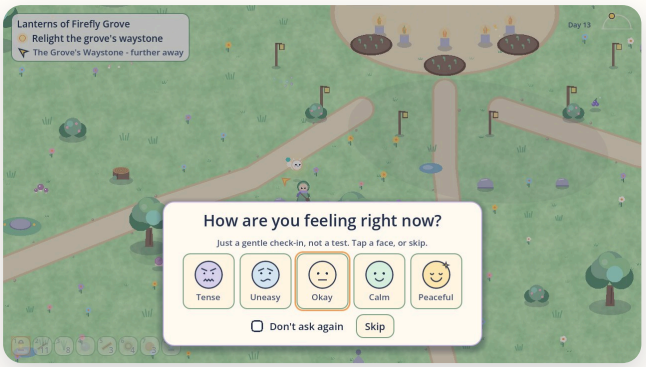
A living world

17 creatures to sketch and befriend, day and night, rain, mist, fog and light shafts.

03 · DESIGNED FOR CALM

One game, two layers

Every player gets a gentle, entirely optional well-being layer. Researchers get a hidden Study Mode that runs **exactly the same build**, so what is studied is what players will actually play.



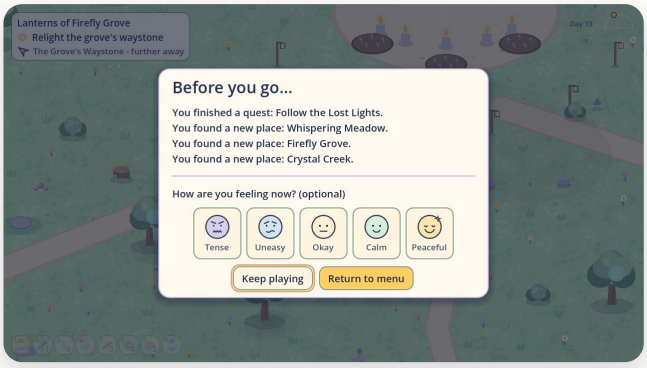
Mood check: five drawn faces, one tap, always skippable, "Don't ask again". Not a test, and not stored unless the player opts in.



Rest spots: 12 quiet places where the player simply sits. The HUD fades, the camera drifts and the music swells.



Breathe with Lumi: Lumi's glow paces slow breathing (4/6, 4/4 or 5/7 s). Any key stops it. No scores, no streaks.



"Before you go...": a warm summary built only from real events of the session, never a score or a grade.

DEFAULT GAME

For every player

- Start Adventure goes straight into the world
- Optional mood check and session summary
- Optional private mood journal, **off by default**
- No research data is ever written

STUDY MODE

For research sessions only

- Hidden: a launch flag or a 7-click unlock with a phrase
- Participant codes, fixed 10 / 15 / 20-minute sessions
- A fresh, identical world for every participant
- Consent-gated CSV files for analysis

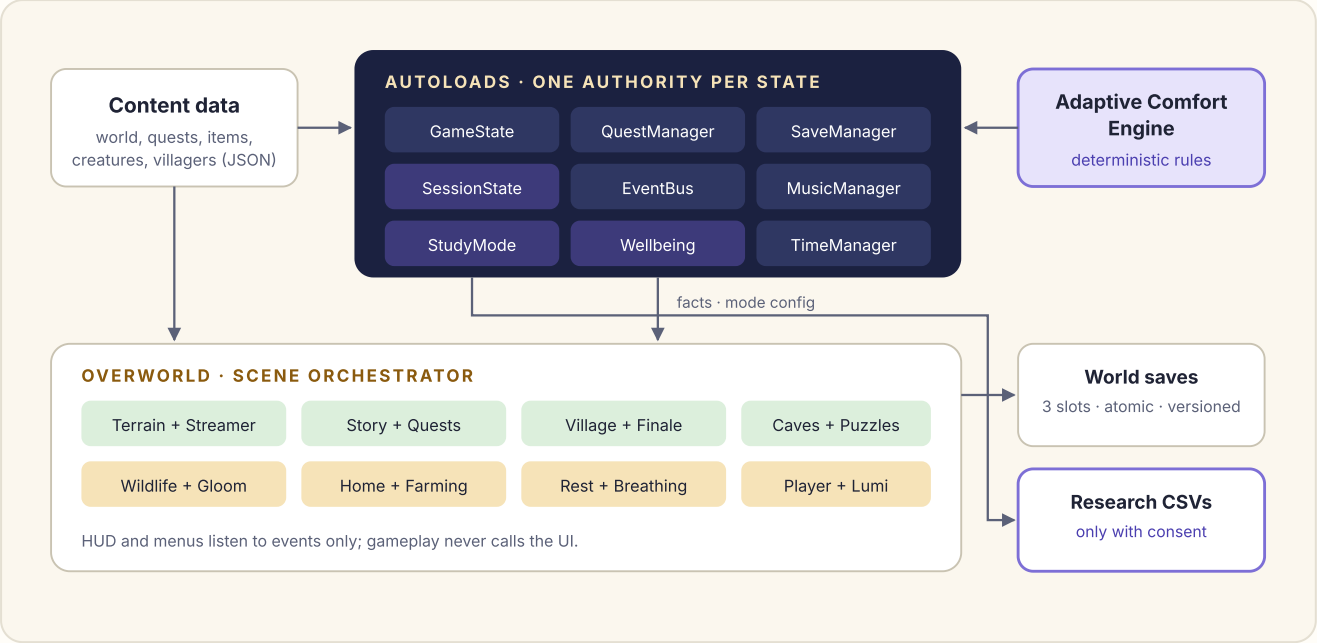
Rules for every well-being feature

Optional, skippable and switchable off in Settings. Never a score, a streak or a comparison. Never a claim that the game reduces stress. Never an inference of the player's emotional state from how they play. Every new screen is translated into Bahasa Indonesia and respects *Reduce motion*.

04 · SYSTEM DESIGN

Engineered like a product, tested like research software

One authority per kind of state, pure logic separated from scenes, and gameplay code that never talks to the UI directly. Content is data, so a new fish, quest or decoration needs no code.



A Deterministic world

A macro map of 40 × 24 authored cells is filled procedurally. Each 1,024 px chunk is seeded by `hash(world_seed, x, y)`, so saves store only **what changed**. At most one chunk is built per frame.

B Quests as a fixed point

Quest conditions read facts from the world state and are re-evaluated until nothing changes. Anything done early still counts, and completed quests can never revert, so players cannot get stuck.

C Robust persistence

Versioned saves with step-by-step migration, atomic writes with a backup, and autosaves on sleep, doors, quests and a timer. Research data is never part of a save.

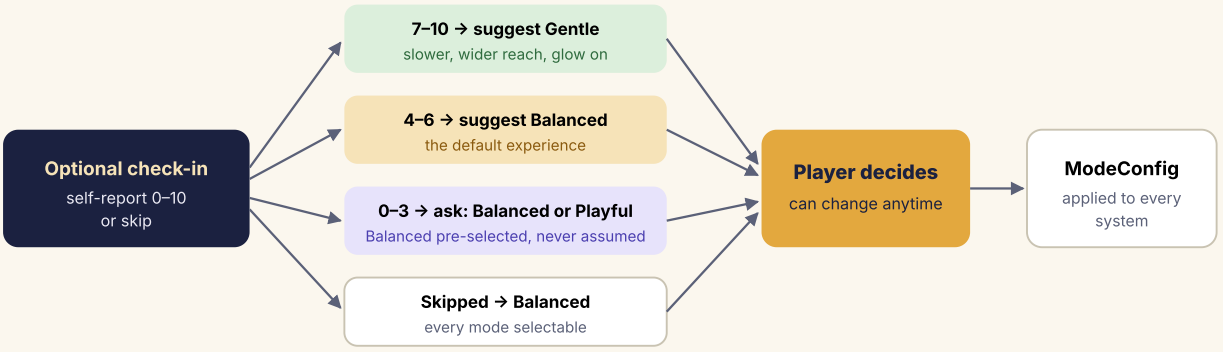
Technology

LAYER	TOOLS AND TECHNIQUES
Engine & language	Godot 4.7.2, statically typed GDScript, GL Compatibility renderer
Art & audio	Procedural drawing and shaders (biome blending, rivers, gloom veil); original MIDI scores composed in Python; synthesized SFX; adaptive crossfading music
Data & analysis	RFC 4180 CSV layer, Python 3.12, pandas, matplotlib, Jupyter; synthetic sample data for reproducibility

05 · EXPLAINABLE ADAPTATION

An "AI" that is a glass box, not a black box

The Adaptive Comfort Engine is a deterministic module with **no machine-learning model** and no scene dependencies, so every rule can be read, tested and audited. It uses only what the player does on purpose.



The engine never raises difficulty automatically and never treats speed, pauses or mistakes as signs of stress.

Every parameter is named and bounded

PARAMETER	GENTLE	BALANCED	PLAYFUL	BOUNDS
Interaction reach (px)	104	72	64	40–140
Light movement ×	0.35	1.0	1.5	0–2
Guiding glow	on	off	off	–
Bonus Prism Motes	0	0	3	0–3
Lantern Catch lanterns	6	8	10 (+2)	3–12
Lantern drift (px/s)	6	18	30	0–40

Helpful, never intrusive

- **3 explicit hint requests** turn on the guiding glow, unless the player turned it off
- **2 presses just outside reach** explain the control and show the reach circle
- **"Calmer" or "livelier"** switches the mode immediately, keeping all progress
- Modes reach every system: hints, fishing (Gentle catches automatically), creatures, puzzles, Lantern Night and the clock

Why rule-based?

With no player data yet, an ML model would be untestable and unexplainable. Rules form a transparent baseline that any future personalization model must beat.

Privacy you can verify, research you can reproduce

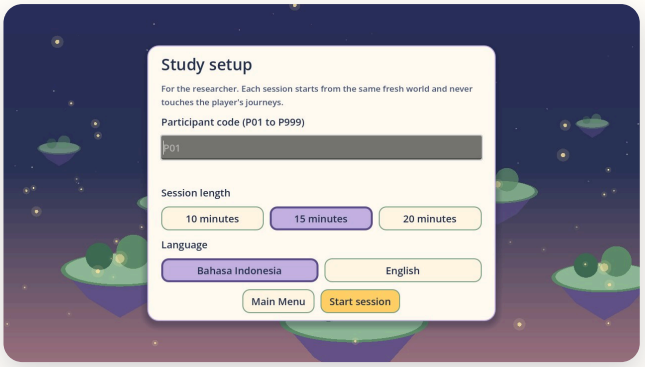
Research data exists only in Study Mode, only after consent, and never leaves the device. A fixed schema, a synthetic sample and tested Python scripts make every analysis reproducible.

Privacy by design

PRINCIPLE	IMPLEMENTATION
Nothing by default	The default game stores no research data
Consent first	Rows are written only after an unticked-by-default checkbox
Nothing identifying	No name, email, account, location or device id
Stays on device	No network code, analytics or trackers
Player control	Export, open folder, delete and reset in-game

Honest statistics

- `score_delta = post − pre`; a skipped answer is an empty cell, never 0
- Invalid rows are excluded whole and reported with a reason
- Neutral, non-causal wording in every summary



Study setup: participant code, session length and language.

```
sessions.csv (schema v1)
schema_version, session_id, session_started_at,
pre_stress_score, recommended_mode, selected_mode,
final_mode, session_duration_seconds, hints_used,
interaction_help_events, items_collected,
garden_restored, post_stress_score, score_delta,
experience_rating, completed

study_sessions.csv participant_code, language,
build_version, target_duration + all v1 columns
study_events.csv coarse activity events
```

Automated quality suite · one command, 18 checks

188 unit tests on pure logic

678 smoke checks, reachability flood-fill

396 overworld checks: plays Acts 1–3, saves, relaunches

175 UI checks in both languages

30 Python & music-tool tests

Also verified: static resource references, all 9 screens load, full Indonesian catalog coverage with matching placeholders. Performance: about 61 fps on average on the development laptop (RTX 3050), with a light budget of 16 dynamic lights and view culling.

07 · PLANNED EVALUATION

How the effect will be tested

CALMQUEST makes **no claim yet** that it reduces stress. That claim must be earned with people. The protocol below defines how, before any data is collected.

Research question

Does a 15-minute CALMQUEST session reduce **state anxiety** in working adults more than a neutral control activity of equal length?

Primary hypothesis

H1: the decrease in state anxiety (STAI-S) from T1 to T2 is larger after CALMQUEST than after the control.

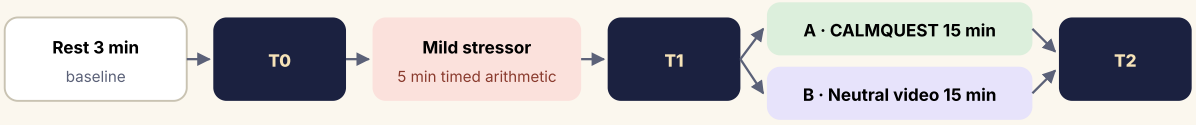
Design

Randomized, counterbalanced crossover (AB/BA), n = 30 working adults aged 18+. Each participant completes both conditions on separate days at the same time of day. Each person serves as their own control.

Why crossover

Power ≈ 0.75 for a medium effect (dz = 0.5, α = .05), versus ≈ 0.26 for two independent groups of 15.

Session procedure (≈ 45 minutes)



Day 1 adds consent, demographics and PSS-10. After the game condition: GEQ, SUS and a short interview, then debriefing.

Instruments

INSTRUMENT	MEASURES	WHEN	ROLE
STAI-S (State-Trait Anxiety Inventory, state form)	State anxiety	T0, T1, T2	PRIMARY
SAM (Self-Assessment Manikin)	Emotional valence and arousal	T0, T1, T2	Secondary
PANAS	Positive and negative affect	T1, T2	Secondary
In-game 0–10 self-report	Subjective stress	T1, T2	Secondary
PSS-10, Indonesian version (Hakim et al., 2024)	Perceived stress, past month	Day 1	Sample description
GEQ core & post-game; **SUS**	Game experience; usability	After game	Descriptive

Analysis

2 × 2 repeated-measures model (condition × time), or a linear mixed model with period and sequence effects; Wilcoxon signed-rank as a fallback. Effect sizes (Cohen's dz) with 95% confidence intervals, not p-values alone. Exploratory: link Study Mode activity logs to score changes.

Ethics & transparency

Ethical clearance from a university research ethics committee, informed consent, anonymous participant codes, the right to withdraw at any time, debriefing, and pre-registration on OSF before data collection. Exclusion: current treatment for anxiety or depression, photosensitive epilepsy.

08 · ROADMAP

From prototype to public release

Development is complete. The next milestones move CALMQUEST from automated verification to evidence with real people, and then to players worldwide.



Dates are planned estimates. Study dates depend on ethical clearance.

Beyond the first study

- Orchestral renders of the 14 original MIDI scores (currently synthesized placeholders)
- A preference study: fixed versus player-chosen comfort mode
- Only with enough consented data: a personalization model, evaluated against the rule-based baseline
- More regions, a shared "Dream Garden" co-op mode

Known limitations

- Not yet play-tested by people; verified by automated tests and scripted screenshots
- Art is procedural and intentionally simple; Windows only so far
- The in-game 0–10 item is not a validated instrument, which is why the study uses STAI-S



The finale: the villagers gather as the light returns to Lumeria.

A note on claims. CALMQUEST is an entertainment prototype oriented toward well-being. It is not a medical device and does not diagnose, treat or measure stress or anxiety. Its effect on stress has not been tested yet; the study in Section 07 is how it will be.

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github.com/dkiplikurniawan/calmquest
Godot 4.7 · GDScript · Python 3.12